

# MAT 523: Functions of a Real Variable I

## Homework VII

Nolan R. H. Gagnon

(1.)

### Problem

---

Theorem 15 asserts that if  $E_1 \supset E_2 \supset \cdots$  is a descending chain of measurable sets with  $m(E_1) < \infty$ , then

$$m\left(\bigcap_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} m(E_n).$$

Show by example that the statement need not hold if  $m(E_1) = \infty$ .

### Solution

---

Let  $E_n = (n, \infty)$ . Then  $E_1 \supset E_2 \supset \cdots$  is a descending chain of measurable sets with  $m(E_1) = \ell[(1, \infty)] = \infty$ . As we showed in a previous homework,  $\bigcap_{n=1}^{\infty} E_n = \emptyset$ . Thus,

$$m\left(\bigcap_{n=1}^{\infty} E_n\right) = m(\emptyset) = 0.$$

The measure of any  $E_n$ , however, is

$$m(E_n) = \ell[(n, \infty)] = \infty.$$

Thus,

$$\lim_{n \rightarrow \infty} m(E_n) = \lim_{n \rightarrow \infty} \infty = \infty.$$

We can now see that the statement does not hold for this particular descending chain of sets.

(2.)

**Statement**

---

The Cantor set has no isolated points.

**Proof**

---

Let  $\mathcal{C}$  represent the Cantor set and let  $x \in \mathcal{C}$ . Let  $\varepsilon > 0$ . We will show that the open interval  $(x - \varepsilon, x + \varepsilon)$  contains a point in  $\mathcal{C}$  that is not equal to  $x$ . Fix  $N \in \mathbb{N}$  such that  $\frac{1}{3^N} < \varepsilon$ . Since  $x \in \mathcal{C}$ , it is in each of the  $C_k$  whose intersection forms  $\mathcal{C}$ ; in particular, it is in  $C_N$ . This means it is in one of the disjoint interval components of length  $\frac{1}{3^N}$  that comprise  $C_N$ . Denote this interval by  $I = [a, b]$ . Both  $a$  and  $b$  are elements of  $\mathcal{C}$  (by construction). Since  $b - a = \frac{1}{3^N} < \varepsilon$  and  $x \in [a, b]$ , it must be the case that  $[a, b] \subset (x - \varepsilon, x + \varepsilon)$ . This means both  $a$  and  $b$  are in  $(x - \varepsilon, x + \varepsilon)$ . Thus, there are at least two distinct elements of  $\mathcal{C}$  in  $(x - \varepsilon, x + \varepsilon)$ . Therefore,  $\mathcal{C}$  contains no isolated points.

■

(3.)

**Statement**

---

Between any two elements  $x, y \in \mathcal{C}$ , there is a real number  $t \notin \mathcal{C}$ .

**Proof**

---

Let  $x, y \in \mathcal{C}$ . Fix  $N \in \mathbb{N}$  such that  $\frac{1}{3^N} < |x - y|$ . Now, since  $x, y \in \mathcal{C}$ , they are in each of the  $C_k$  whose intersection forms  $\mathcal{C}$ ; in particular, they are in  $C_N$ . This means they are each in exactly one of the disjoint interval components of length  $\frac{1}{3^N}$  that comprise  $C_N$ . Since  $|x - y| > \frac{1}{3^N}$ , it is not possible for  $x$  and  $y$  to be in the same interval component. Therefore, there is an open interval  $(a, b)$  of length at least  $\frac{1}{3^N}$  such that  $(a, b) \subset [x, y]$  but  $(a, b) \not\subset \mathcal{C}$ . Since  $(a, b)$  has nonzero length, it contains a real number  $t \notin \mathcal{C}$  (in fact, it contains uncountably many such numbers).

■

(4.)

**Statement**

---

Fix a number  $\alpha \in (0, 1)$ . Let  $\mathcal{F}$  be the subset of  $[0, 1]$  constructed in the same manner as the Cantor set except that each of the intervals removed at the  $n$ th deletion stage has length  $\alpha 3^{-n}$ . Show that  $\mathcal{F}$  is a closed set,  $[0, 1] \setminus \mathcal{F}$  is dense in  $[0, 1]$ , and  $m(\mathcal{F}) = 1 - \alpha$ .

**Proof**

---

Since  $\mathcal{F}$  is constructed in the same manner as  $\mathcal{C}$ , it is an intersection of a collection of closed sets. Therefore, it is closed.

To show that  $[0, 1] \setminus \mathcal{F}$  is dense in  $[0, 1]$ , we begin by showing that  $\mathcal{F}$  is nowhere dense in  $[0, 1]$ . For each  $n \in \mathbb{N}$ , let  $F_n$  be analogous to  $C_n$  from the previous two problems. Let  $x, y \in \mathcal{F}$ . Denote by  $\ell_n$  the length of the interval components of  $F_n$ . Fix  $N \in \mathbb{N}$  such that  $\ell_n < |x - y|$ . By the same reasoning used in Problem 3, it is impossible for  $x$  and  $y$  to be in the same interval component of  $F_N$ . Therefore, there is a nondegenerate open interval  $(a, b)$  such that  $(a, b) \subset [x, y]$  but  $(a, b) \not\subset \mathcal{F}$ . Since  $(a, b)$  is nondegenerate, it contains a real number  $t \notin \mathcal{F}$ . Thus,  $\mathcal{F}$  is nowhere dense. Now, let  $c, d \in [0, 1]$ . Suppose  $c, d \in \mathcal{F}$ . Then since  $\mathcal{F}$  is nowhere dense, there is an element  $g \notin \mathcal{F}$  (i.e.,  $g \in [0, 1] \setminus \mathcal{F}$ ) such that  $c < g < d$ . Next suppose  $c, d \in [0, 1] \setminus \mathcal{F}$ . (Proof not complete)

The measure of  $[0, 1]$  is 1. Each of the subsets removed from  $[0, 1]$  in the creation of  $\mathcal{F}$  is an interval, so the measure of each of these subsets is just its length. Since  $[0, 1]$  is measurable and each of the deleted intervals is measurable, we can use the Excision Property to find the measure of  $\mathcal{F}$ . Now, on the  $k$ th stage of the creation of  $\mathcal{F}$ , a total length of  $\alpha \left(\frac{2}{3}\right)^k$  is removed from  $[0, 1]$ . Thus, in the entire deletion process,

we remove a length of

$$\begin{aligned}
 \sum_{k=1}^{\infty} \alpha \left( \frac{2}{3} \right)^k &= \frac{\alpha}{3} \sum_{k=1}^{\infty} \left( \frac{2}{3} \right)^{k-1} \\
 &= \frac{\alpha}{3} \sum_{j=0}^{\infty} \left( \frac{2}{3} \right)^j \\
 &= \frac{\alpha}{3} \left[ \frac{1}{1 - \frac{2}{3}} \right] \\
 &= \frac{\alpha}{3} \left[ \frac{1}{\frac{1}{3}} \right] \\
 &= \alpha.
 \end{aligned}$$

Therefore, by the Excision Property, the measure of  $\mathcal{F}$  is  $1 - \alpha$ .

■